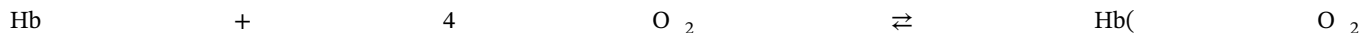


The reversible reaction



exists in blood between hemoglobin (Hb) and oxyhemoglobin $\text{Hb}(\text{O}_2)_4$. According to the law of mass action, doubling the O_2 concentration would:

- ☐ A. decrease the reaction quotient by a factor of 2. [10%]
- ✓ ☒ B. decrease the reaction quotient by a factor of 16. [57%]
- ☐ C. increase the reaction quotient by a factor of 2. [17%]
- ☐ D. increase the reaction quotient by a factor of 8. [15%]

Correct

57% answered correctly

Explanation:

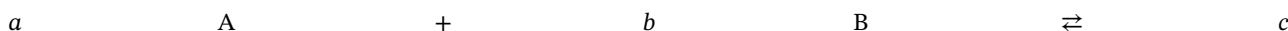
Effect on reaction quotient Q when $[\text{O}_2]$ is doubled

$$\text{Hb} + 4\text{O}_2 \rightleftharpoons \text{Hb}(\text{O}_2)_4$$

$$[\text{O}_2] = x \quad Q = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}][\text{O}_2]^4} = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}]x^4} = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}]x^4}$$

$$[\text{O}_2] = 2x \quad Q = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}][2x]^4} = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}]16x^4} = \frac{1}{16} \left(\frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}]x^4} \right)$$

The general reaction in which chemical species A and B react to form products C and D is written as



where a , b , c , and d are the coefficients of the balanced reaction. The **law of mass action** states that a **reaction quotient Q** can be expressed as a **ratio** of the molar concentrations of the products over the reactants, each raised to the power of its respective balanced **reaction coefficient**:

$$Q = \frac{[\text{products}]^{\text{coeff}}}{[\text{reactants}]^{\text{coeff}}}$$

The law of mass action further states that the concentrations of the chemical species will **change** until **equilibrium** is reached, such that Q and the equilibrium constant K_{eq} are equal. Changes to the concentration of chemical species with a higher reaction coefficient will have the greatest impact to increase or decrease the value of Q .

For the reaction between hemoglobin and oxygen, Q is expressed as

$$Q = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}][\text{O}_2]^4}$$

If $[\text{O}_2]$ is initially x mol/L and is then doubled to $2x$ mol/L, the change will be raised to the power of 4 ($[2x]^4 = 16x^4$). Because $[\text{O}_2]$ is in the denominator of the Q expression, this will decrease the value of the reaction quotient by a factor of 16 relative to the initial value of Q .

(Choice A) The reaction quotient would decrease by half if the *hemoglobin* concentration were doubled instead of the O_2 concentration.

(Choice C) Doubling the *oxyhemoglobin* concentration instead of the O_2 concentration would increase the reaction quotient by a factor of 2.

(Choice D) O_2 has a coefficient of 4 in the balanced reaction equation. In the reaction quotient, $[\text{O}_2]$ is in the denominator and the concentration of O_2 is exponentially raised to the power of 4 (not just multiplied by 4).

Educational objective:

The law of mass action states that a reaction quotient Q can be expressed as a ratio of the molar concentrations of the products over the reactants, each raised to the power of its respective balanced reaction coefficient. Reactions tend to proceed toward equilibrium by shifting species concentrations until Q equals the equilibrium constant K_{eq} .